

5S Material Allocation Process Mapping and Business Requirements Document

Portfolio Sample Work - Jindal Stainless Ltd

Steel Melt Shop | AOD Operations

Field	Details
Document Type	Process Mapping and Business Requirements Document
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Role Context	Operations Analyst performing Business Analysis activities
Business Area	AOD material allocation, visual management and 5S
Method Used	DMAIC with BA documentation structure
Purpose	To show requirement gathering, process mapping, gap analysis, business rules and process improvement skills from Jindal work experience

1. Project Background

This project focused on 5S implementation for material allocation at the AOD area in the Steel Melt Shop. The original problem statement was to complete 5S of material allocation at AOD to improve visual management and reduce material wastage. The work included process review, stakeholder discussion, material location tagging, bin allocation, receipt handover control and improvement of material handling practices.

From a Business Analyst perspective, the project required understanding the current operating process, identifying pain points, capturing stakeholder requirements, defining business rules and documenting the future process for better shop-floor execution.

2. Business Problem

Problem Area	Business Problem
Material receipt delay	Material receipt information could move from one shift to another, creating confusion during handover.
Visual management	Material locations were not always easy to identify from the control room or shop floor.
Space utilisation	Long-stored materials and unplanned placement reduced available working space.
Small quantity material handling	Materials such as FeMo, Mn metal, pure Ni and pure Cu required better sub-bin allocation.
Communication gap	Melters, operators, technicians and helpers needed a clear location reference for each material.
Grade-wise equipment readiness	Number of chutes required varied by grade family, so availability needed to match the running heat plan.
Ownership gap	Material unloading and placement required clear responsibility to avoid wrong placement, especially during night shift operations.

3. Business Objectives

Objective	Description
Improve visual management	Create clear identification and tagging for material locations.
Reduce material wastage	Improve material allocation discipline and reduce wrong placement.
Improve shift handover	Ensure receipt and material information is available before the next shift takes over.
Improve material handling	Define better bins, sub-bins and standard locations for frequently used materials.

Objective	Description
Improve communication	Allow the control room to communicate material location clearly to technicians and helpers.
Improve process control	Align material booking and chute availability with the heat plan and grade requirement.
Sustain 5S	Create controls so that the improved process can be maintained over time.

4. Scope

In Scope	Out of Scope
Material allocation at AOD area	Supplier commercial negotiations
Tagging and nomenclature for material locations	Change in metallurgical grade design
Bin and sub-bin division for small quantity materials	Major civil construction work
Receipt handover and material identification control	ERP or software implementation
Material booking alignment with heat plan	Automation of weighing bridge system
Chute availability by grade requirement	Full plant-wide 5S implementation outside AOD

5. Stakeholder Analysis

Stakeholder	Need / Interest	BA Engagement
Production Manager	Improved productivity and reduced wastage	Understand business priority and success measures
Shift-In-Charge	Clear material availability and handover information	Capture shift-level pain points and execution needs
Melter / Operator	Quick identification of required material	Understand real user workflow from control room
Technician / Helper	Clear physical location of materials	Capture location communication issues
Store / Bunker Yard Team	Correct unloading and allocation reference	Define receipt and placement coordination
Material Unloading Supervisor	Ownership of unloading and correct placement	Define responsibility and handover control
Maintenance / Mechanical Team	Space allocation and equipment placement	Capture constraints related to shop-floor movement and space

6. Methodology Used - DMAIC Mapped to BA Work

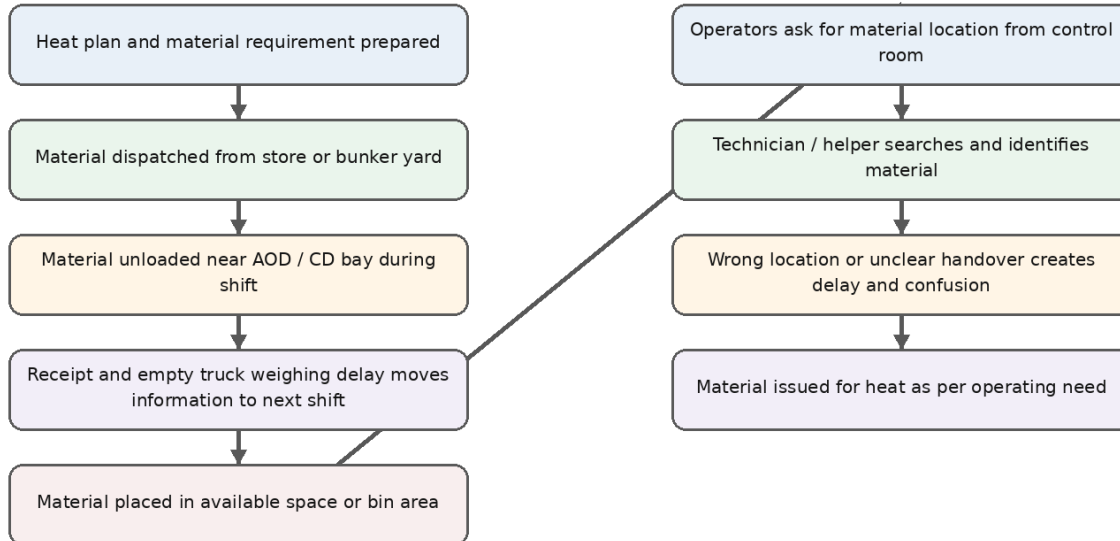
DMAIC Phase	Operations Activity	BA Skill Demonstrated
Define	Defined the need to perform 5S of material allocation at AOD to improve visual management and reduce wastage.	Problem definition, business need, scope and objective setting
Measure	Identified need for nomenclature, bin division and material allocation mechanism.	Current state review, data points and process observation
Analyze	Reviewed current trends and shop-floor issues affecting material allocation.	Gap analysis, root cause thinking and stakeholder discussion
Improve	Proposed material handling improvement, tagging, sub-bins and receipt control.	To-Be process design, business requirements and solution options
Control	Defined ways to implement and maintain 5S at AOD.	Control plan, ownership, KPIs and process governance

7. As-Is Process Map

The As-Is process below represents the material allocation flow before 5S standardisation.

As-Is Process Map: AOD Material Allocation

Current process view before 5S standardisation



8. As-Is Pain Points and Gap Analysis

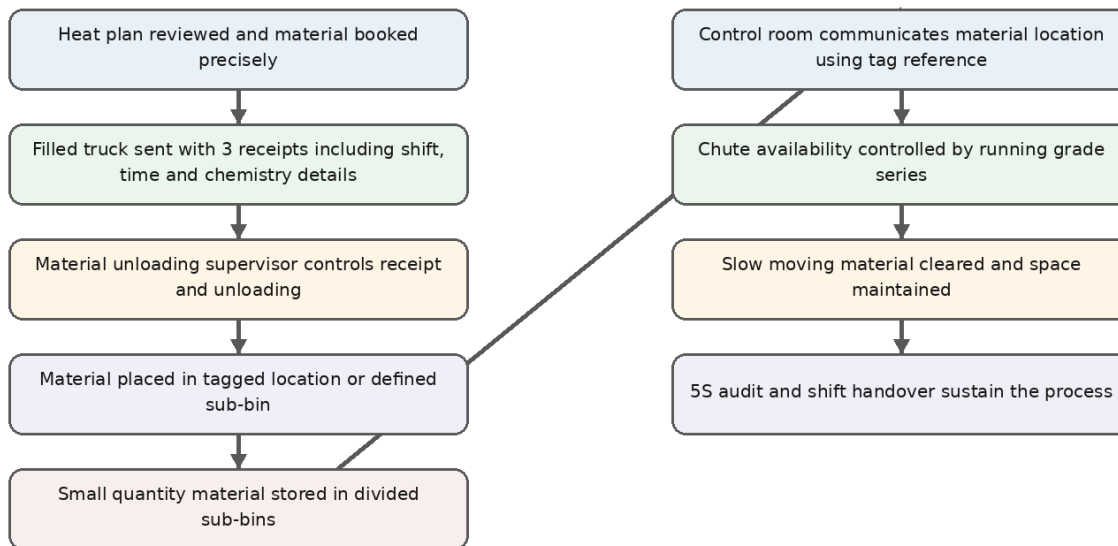
Pain Point	Operational Impact	Required Change
Receipt handover delay	A-Shift to B-Shift information delay created confusion about material receipt.	Use 3 receipts and include shift, timing and chemistry details.
Unclear location identification	Operators and helpers could not identify material locations quickly.	Tag all material locations with high plate welding or permanent labels.
No sub-bin system for small materials	Small quantity materials were harder to locate and control.	Divide bins into sub-bins for FeMo, Mn metal, pure Ni, pure Cu and similar materials.
Material kept for long time	Space remained blocked by slow moving material.	Clear long-stored material gradually and set maximum quantity limits where needed.
Material booking not precise	Extra or wrong material availability affected space and planning.	Book material from stores and bunker yard according to the heat plan.
Chute availability not grade aligned	Unnecessary chutes occupied space, while required chutes could be unavailable.	Vary chute availability based on running grade family.
Wrong material placement	Material placed at wrong location caused execution issues.	Assign material unloading supervisor with material knowledge.

9. To-Be Process Map

The To-Be process introduces clear tagging, defined receipt control, material supervisor ownership and grade-wise allocation discipline.

To-Be Process Map: AOD Material Allocation

Target process view after 5S and visual management controls



10. Business Requirements

Req ID	Business Requirement	Priority
BR-01	The AOD material allocation area must have clearly tagged locations for each identified material category.	High
BR-02	Material bins must be divided into sub-bins for small quantity materials such as FeMo, Mn metal, pure Ni and pure Cu.	High
BR-03	Filled material trucks must carry 3 receipts instead of 2 receipts to support shift handover.	High
BR-04	Receipt details must include shift, timing and chemistry details such as carbon and silicon percentage where applicable.	High
BR-05	Material must be booked from stores and bunker yard precisely according to the heat plan.	High
BR-06	Chute availability must be adjusted based on the running grade family, for example 3 chutes for 300 series and 2 chutes for 400 series.	Medium
BR-07	Slow moving materials such as Mn ore and Al-Bronze must be gradually cleared to release space.	Medium
BR-08	LPFeCr quantity in the area must be controlled to avoid unnecessary space blockage.	Medium
BR-09	Standard weights used for weighing bridge calibration must be tagged.	Medium
BR-10	A material unloading supervisor must be available in CD bay to ensure correct placement and reduce wrong-location issues.	High

11. Functional Requirements

FR ID	Functional Requirement
FR-01	The material allocation process must support location identification from the control room using tagged location references.
FR-02	The process must allow operators to communicate material location to technicians and helpers without ambiguity.
FR-03	The bin structure must support separate storage of small quantity materials.
FR-04	The receipt handover process must ensure that material receipt status is available before shift handover.
FR-05	The material unloading process must include verification by a responsible person before final placement.
FR-06	The process must support grade-wise chute planning based on the running heat plan.

12. Business Rules

Rule ID	Business Rule
BU-01	Each material category must be stored only in its identified tagged location.
BU-02	Small quantity materials must be placed in defined sub-bins and not mixed with bulk materials.
BU-03	Material receipt handover must include receipt count, shift, unloading time and chemistry information where applicable.
BU-04	Material must be booked according to the heat plan rather than excess availability.
BU-05	If 300 series is running, required chutes should be available as per grade plan.
BU-06	If 400 series is running, chute requirement must be adjusted to avoid unnecessary space usage.
BU-07	Long-stored materials must be reviewed and cleared gradually to create space.
BU-08	The material unloading supervisor must ensure material is not placed at the wrong location.
BU-09	Tagging must be visible enough for store, bunker yard and AOD team members to identify locations.

13. Acceptance Criteria

AC ID	Acceptance Criteria
AC-01	Given a material truck is unloaded, when shift handover happens, then the next shift must receive clear receipt information without confusion.
AC-02	Given a melter or operator asks for material location, when the control room checks the tagged layout, then the location can be communicated clearly to the technician or helper.
AC-03	Given small quantity materials are received, when they are stored, then they must be placed in the correct sub-bin.
AC-04	Given a heat plan is available, when material is booked from stores or bunker yard, then booking must match the heat plan requirement.
AC-05	Given a grade family is running, when chutes are arranged, then only the required number of chutes should occupy the area.
AC-06	Given a 5S audit is performed, when the material area is reviewed, then material tags, bin allocation and space clearance should be visible and traceable.

14. RACI Matrix

Activity	Operations Analyst / BA	Shift-In-Charge	Store / Bunker Yard	Unloading Supervisor	Production Manager
Capture current process and pain points	R	C	C	C	A
Define material tagging requirement	R	C	C	C	A
Prepare bin and sub-bin allocation logic	R	C	C	C	A
Implement receipt handover control	C	A	R	R	C
Ensure correct material placement	C	A	C	R	C
Monitor 5S sustainment	C	R	C	R	A

15. KPI and Success Measures

KPI	Measurement Method	Target / Expected Result
Material location identification time	Time taken to identify material location from control room or shop floor	Reduced search time
Wrong placement incidents	Count of materials found in incorrect location	Reduced incidents
Receipt handover accuracy	Receipt count and information available during shift handover	Improved handover accuracy
Tagged location completion	Number of material locations with visible tags	100 percent critical locations tagged
Sub-bin readiness	Availability of sub-bins for small quantity materials	Defined sub-bins available
Space availability	Review of blocked space due to slow moving materials	Improved space usage
5S compliance score	Periodic 5S audit of AOD material allocation area	Sustained compliance

16. Benefits Delivered / Expected

Benefit Area	Benefit
Visual management	Material locations become easier to identify for AOD team, store and bunker yard.
Communication	Melter or operator can guide technician and helper using clear location references.
Shift handover	Receipt information becomes clearer between shifts.
Space management	Long-stored and excess material can be controlled to improve working space.
Material control	Small quantity materials are separated and easier to track.
Operational efficiency	Less time is lost in searching, clarifying and correcting material placement.
Sustainability	Control measures help maintain 5S after implementation.

17. Final Summary

The 5S Material Allocation project at AOD is a strong example of operations-based Business Analysis work. The project focused on improving visual management, reducing material wastage, strengthening shift handover, improving material location clarity and sustaining better shop-floor discipline.

By presenting the work as a Process Mapping and Business Requirements Document, it clearly demonstrates BA skills such as requirement gathering, stakeholder coordination, As-Is and To-Be process mapping, gap analysis, business rules, acceptance criteria, RACI and KPI-based improvement.